Hospital Appointment Booking System



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OVERVIEW:

- Introduction
- Objectives
- Key Features
- Modules & Functionalities
- Flowchart
- Technologies Used
- Conclusion

INTRODUCTION:

In today’s fast-paced world, hospitals require efficient systems to manage appointments without delays or manual errors. This project presents a console-based Hospital Appointment Booking System in C++, allowing patients to book, view, and cancel appointments, doctors to manage their schedules, and administrators to oversee all activities. Designed to simulate real-world hospital operations, it demonstrates structured data handling, user-friendly interaction, and practical problem-solving through programming.



OBJECTIVES:

- Apply programming fundamentals in a real project
- Use structures, arrays, vectors, and modular functions
- Implement user authentication and role-based access
- Simulate a real-life hospital system
- Handle multiple entities (patients, doctors, appointments)
- Bridge the gap between theory and practical application





KEY FEATURES:

- Add doctor details (ID, name, specialization, availability)
- Book appointments by doctor availability
- View all booked appointments
- Cancel appointments
- User-friendly console interface
- Menu-driven navigation



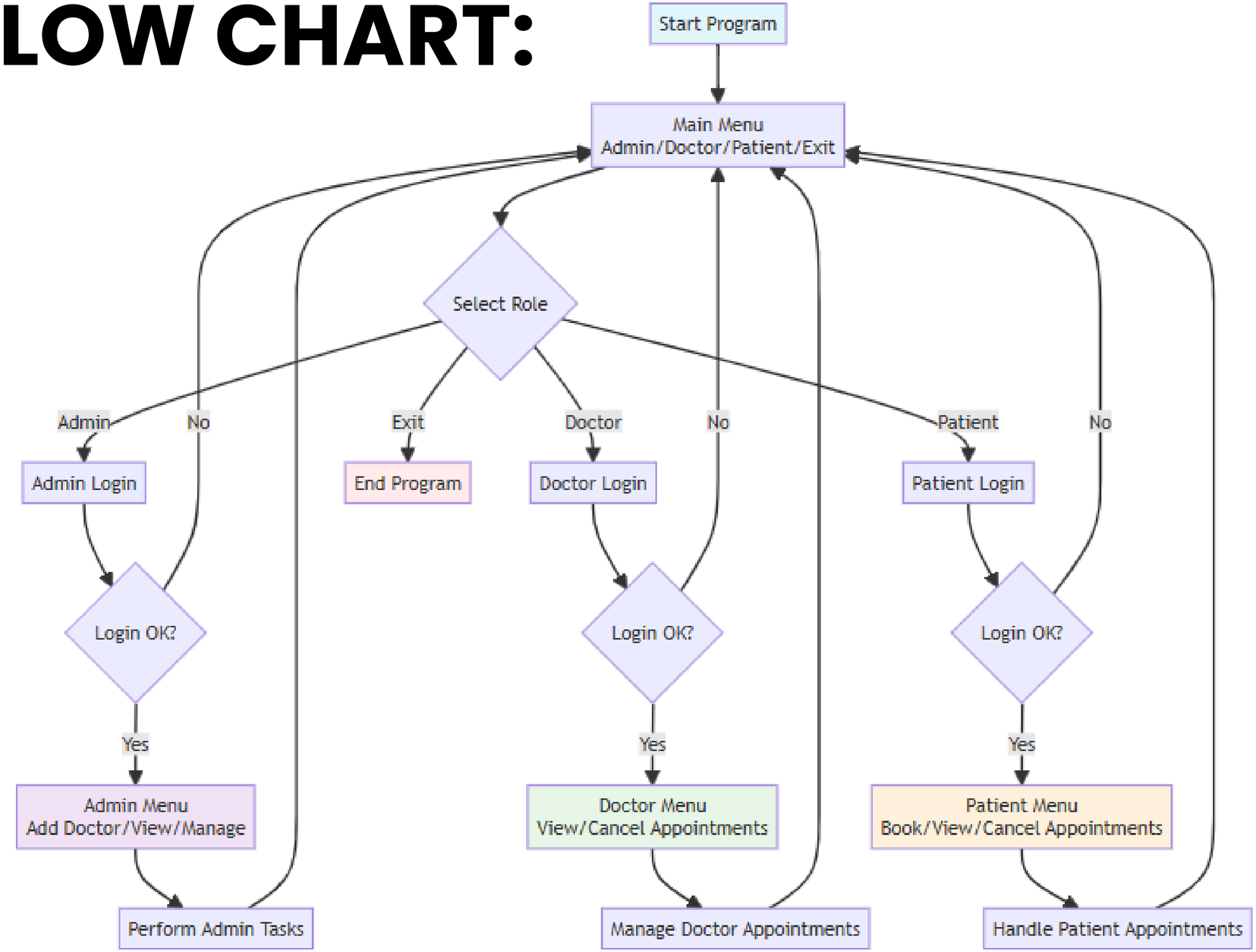
MODULES & FUNCTIONALITIES:

- **Login Module**: Validates user credentials and grants role-based access
- **Booking Module**: Patients book appointments by doctor ID and date
- **Viewing Module**: Users view appointments based on role
- **Cancellation Module**: Patients/doctors cancel appointments
- **Doctor Management**: Admins add doctor info (ID, name, specialization, schedule)
- **Patient Registration:** Admins register patients with username and password





FLOW CHART:


Start Program
Main Menu
Admin/Doctor/Patient/Exit
Select Role
Admin
No
Exit
Doctor
No
Patient
No
Admin Login
End Program
Doctor Login
Patient Login
Login OK?
Login OK?
Login OK?
Yes
Yes
Yes
Admin Menu
Add Doctor/View/Manage
Doctor Menu
View/Cancel Appointments
Patient Menu
Book/View/Cancel Appointments
Perform Admin Tasks
Manage Doctor Appointments
Handle Patient Appointments



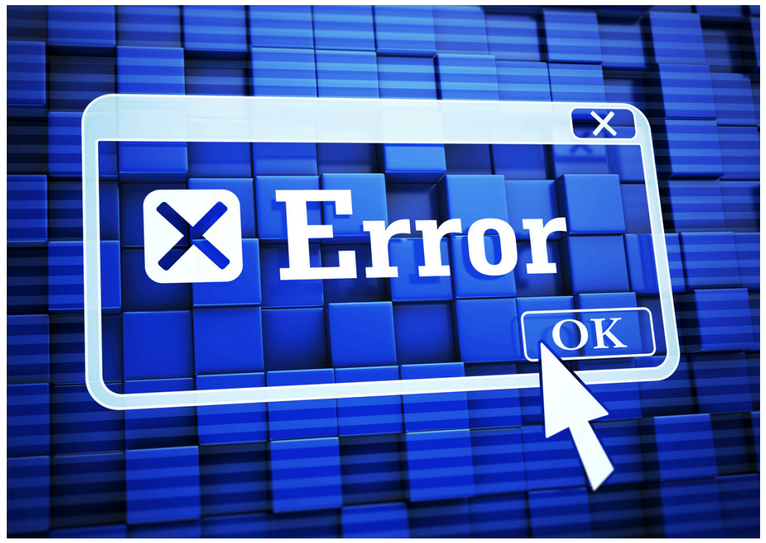


FUTURE SCOPE & ENHANCEMENTS:

- Waitlist for fully booked slots
- Telemedicine for remote appointments
- Advanced doctor availability filters
- Appointment reminders and notifications

Conclusion

The Hospital Appointment Booking System demonstrates how technology can streamline and improve the healthcare process. By automating appointment scheduling, we not only reduced administrative workload but also provided patients with an easy, accessible way to manage their healthcare appointments. This project helped us deepen our understanding of hospital management workflows and the complexities involved in developing a secure, reliable, and user-friendly application.





Thank You for

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